Tropical Hypersurfaces and Gradient Interpretability in ReLU Networks

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Abstract

This report follows Option 1 and is based on the paper of Zhang, Naitzat, and Lim, *Tropical Geometry of Deep Neural Networks* [1]. I study ReLU networks with integer weights from a tropical viewpoint and show how their tropical hypersurfaces encode the piecewise-constant gradient field. In particular, I analyze a shallow ReLU network, prove that gradient discontinuities occur exactly along the hidden tropical hypersurfaces, and verify this numerically on a two-dimensional example.

1 Introduction

A central theme in this course was to study mappings through their Jacobians. For a parameterized model

$$F(\theta) : \mathbb{R}^k \to \mathbb{R}^m,$$

the Jacobian

$$J_F(\theta) = \frac{\partial F}{\partial \theta}$$

describes how the output responds to changes in the parameters. For neural networks, however, an equally important object is the Jacobian with respect to the input. When the network produces a scalar output, this reduces to the gradient

$$\nabla_x \nu(x) \in \mathbb{R}^d,$$

which records the instantaneous sensitivity of the prediction to changes in each input coordinate. Understanding how this gradient behaves, and, crucially, where it changes, is therefore a natural interpretability question.

In the paper of Zhang, Naitzat, and Lim [1], the main emphasis is on the expressivity of ReLU networks: with integer weights, such networks compute tropical rational functions, and their linear regions and decision boundaries correspond to tropical hypersurfaces arising from Newton polytopes. In this project I take the same tropical viewpoint but shift the focus to interpretability, asking: How is the gradient field of a ReLU network encoded in its associated tropical hypersurfaces, and what does this reveal about the model itself?

2 Background

ReLU networks and integer weights

A fully connected feed-forward ReLU network with L layers and input dimension d is a map

$$\nu \colon \mathbb{R}^d \to \mathbb{R}^p$$

of the form

$$x^{(0)} = x, \qquad x^{(l)} = \sigma\big(W^{(l)} x^{(l-1)} + b^{(l)}\big) \quad (1 \le l \le L-1),$$
$$\nu(x) = W^{(L)} x^{(L-1)} + b^{(L)},$$

where $\sigma(z) = \max\{z, 0\}$ is applied on each coordinate. The parameters are the weight matrices and bias vectors $W^{(l)}, b^{(l)}$. Following Zhang, Naitzat, and Lim [1], I focus on the case where the weight matrices $W^{(l)}$ have integer entries.

Tropical polynomials and tropical hypersurfaces

Tropical geometry provides a natural language for describing such piecewise-linear maps. Following Maclagan and Sturmfels [2], the max-plus tropical semiring $(\mathbb{T}, \oplus, \odot)$ is defined by

$$\mathbb{T} = \mathbb{R} \cup \{-\infty\}, \quad a \oplus b = \max\{a, b\}, \quad a \odot b = a + b.$$

The additive identity is $-\infty$, and the multiplicative identity is 0.

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{Z}^d$ and $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$, the tropical monomial with coefficient $c \in \mathbb{R}$ is the function

$$c \odot x^{\alpha} \quad \text{defined by} \quad (c \odot x^{\alpha})(x) = c + \langle \alpha, x \rangle = c + \alpha_1 x_1 + \cdots + \alpha_d x_d.$$

A tropical polynomial is a tropical sum of finitely many monomials,

$$F(x) = \bigoplus_{k=1}^{r} c_k \odot x^{\alpha_k} = \max_{1 \le k \le r} \{c_k + \langle \alpha_k, x \rangle\}, \qquad x \in \mathbb{R}^d.$$

As an ordinary function $F \colon \mathbb{R}^d \to \mathbb{R}$, this is a convex piecewise-linear function obtained as the pointwise maximum of finitely many affine functions [2, Prop. 3.1.1].

The Newton polytope of F is the convex hull of the exponent vectors,

$$\Delta(F) = \operatorname{conv}\{\alpha_1, \ldots, \alpha_r\} \subset \mathbb{R}^d.$$

The coefficients c_k induce a regular subdivision of $\Delta(F)$ as follows. One considers the lifted points $(\alpha_k, c_k) \in \mathbb{R}^{d+1}$ and the convex hull

$$P(F) = \operatorname{conv}\{(\alpha_k, c_k) : k = 1, \ldots, r\} \subset \mathbb{R}^{d+1}.$$

The upper faces of $P(F)$ project to a subdivision $\delta(F)$ of $\Delta(F)$ under the projection $(\alpha, h) \mapsto \alpha$. The complement of the tropical hypersurface

$$\mathcal{T}(F) = \{x \in \mathbb{R}^d : \text{the maximum in } F(x) \text{ is attained by at least two terms}\}$$

is partitioned into open polyhedra that are dual to the cells in $\delta(F)$.

ReLU networks as tropical rational maps

A tropical rational function is a difference $R = F - G$ of two tropical polynomials. Since F and G are convex piecewise-linear, R is a continuous piecewise-linear function, but not necessarily convex. The work of Zhang, Naitzat, and Lim [1] shows that ReLU networks with integer weights are exactly tropical rational functions.

For simplicity consider a scalar-valued network $\nu \colon \mathbb{R}^d \to \mathbb{R}$ with integer weight matrices $W^{(l)} \in \mathbb{Z}^{n_l \times n_{l-1}}$ and real biases $b^{(l)}$. Zhang, Naitzat, and Lim prove that each coordinate of $\nu(x)$ can be written in the form

$$\nu(x) = F(x) - G(x),$$

where F and G are tropical polynomials whose exponent vectors are integer-valued [1, Thm. 3.4]. The proof proceeds by induction on the depth L. The key idea is to decompose any weight matrix $W^{(l)}$ as $W_+^{(l)} - W_-^{(l)}$ with $W_\pm^{(l)} \in \mathbb{Z}_{\ge 0}^{n_l \times n_{l-1}}$ and then express the linear combination of tropical rationals of previous layers as a tropical rational again. The ReLU nonlinearity $\sigma(z) = \max\{z, 0\}$ can be interpreted as a tropical sum with the constant monomial 0, and this step preserves the tropical rational form.

Conversely, they show that any tropical rational map with integer slopes can be realized by a ReLU network with integer weights and finite depth. Tropical geometry therefore provides an expressive model for the piecewise-linear structure of the network, including its linear regions and decision boundaries.

3 Results

3.1 Gradient fields for a ReLU network

Consider a scalar-valued network with one hidden layer,

$$\nu(x) = w_2^\top \sigma(W_1 x + b_1) + b_2, \qquad x \in \mathbb{R}^d,$$

where $W_1 \in \mathbb{Z}^{n_1 \times d}$, $b_1 \in \mathbb{R}^{n_1}$, $w_2 \in \mathbb{Z}^{n_1}$, and $b_2 \in \mathbb{R}$. Let $h(x) = W_1 x + b_1$ denote the vector of preactivations in the hidden layer, and define the activation pattern

$$m(x) = \big(m_1(x), \dots, m_{n_1}(x)\big), \qquad m_i(x) = 1_{h_i(x)>0}.$$

The hyperplanes

$$H_i = \{x \in \mathbb{R}^d : h_i(x) = 0\}$$

are exactly where the ith hidden unit changes its activation state. We can therefore make the following observation.

Statement 3.1. *The gradient $\nabla\nu(x)$ exists and is constant on each connected component of the complement of*

$$\bigcup_{i=1}^{n_1} H_i,$$

and the set of points where $\nabla\nu(x)$ fails to be continuous is contained in $\bigcup_i H_i$.

Proof. On any region R contained in $\mathbb{R}^d \setminus \bigcup_i H_i$, each $h_i(x)$ retains a fixed sign, so the activation pattern $m(x)$ is constant on R and equal to some $m \in \{0,1\}^{n_1}$. On that region we can write

$$\sigma(h(x)) = \operatorname{diag}(m) h(x),$$

and therefore

$$\nu(x) = w_2^\top \operatorname{diag}(m)(W_1 x + b_1) + b_2 = \underbrace{w_2^\top \operatorname{diag}(m) W_1}_{=a_m^\top} x + \underbrace{w_2^\top \operatorname{diag}(m) b_1 + b_2}_{=c_m}.$$

This is an affine function of x on R. Consequently $\nabla\nu(x) = a_m$ is constant on R, and ν is differentiable at every point of R. Since the activation patterns of ν are constant exactly on the connected components of the complement of $\bigcup_i H_i$, the gradient is constant on each such component.

When x crosses one of the hyperplanes H_i, the sign of $h_i(x)$ can change, which can change the activation pattern $m(x)$ and hence the vector a_m. Thus any discontinuity of $\nabla\nu$ must lie in the union of the H_i.

Thus, the set of possible gradient vectors is finite and indexed by the activation patterns $m \in \{0,1\}^{n_1}$. Indeed

$$\nabla\nu(x) = \sum_{i=1}^{n_1} m_i(x)\, w_2(i)\, W_1(i,:),$$

so each hidden unit contributes its weight vector $W_1(i,:)$ scaled by the output weight. When the weights are integers, these gradient vectors are integer points in $\mathbb{Z}^d$.

Continuing under such an assumption, for each hidden unit define

$$\ell_i(x) = W_1(i,:)x + b_1(i),$$

and the hidden output

$$f_i(x) = \sigma(\ell_i(x)) = \max\{\ell_i(x), 0\}.$$

This function can be written as a tropical polynomial with two monomials.

$$f_i(x) = 0 \odot x^0 \ \oplus\ b_1(i) \odot x^{\alpha_i} = \max\{0, b_1(i) + \langle \alpha_i, x\rangle\},$$

where the exponent vector $\alpha_i \in \mathbb{Z}^d$ is the integer vector of coefficients appearing in $\ell_i(x)$. The support of f_i consists of the two exponent vectors 0 and α_i, so the Newton polytope $\Delta(f_i)$ is the line segment between them. The lifted Newton polytope $P(f_i)$ has two vertices $(0,0)$ and $(\alpha_i, b_1(i))$, so its only upper face is the edge joining these points. The tropical hypersurface

$$\mathcal{T}(f_i) = \{x : f_i \text{ non-differentiable at } x\}$$

is precisely the hyperplane where these terms are equal:

$$\mathcal{T}(f_i) = \{x : 0 = b_1(i) + \langle \alpha_i, x\rangle\} = H_i.$$

In dimension $d = 2$ each $\mathcal{T}(f_i)$ is a tropical line, degenerated to a single straight line because there are only two monomials. The arrangement of these lines partitions the plane into convex polygons on which the gradient is constant.

3.2 Numerical example

For a computational experiment I consider a two-dimensional network with four hidden units and integer weights, matching the assumptions of Zhang, Naitzat, and Lim [1]. The network is

$$\nu(x) = W_2\, \sigma(W_1 x + b_1) + b_2, \qquad x = (x_1, x_2) \in \mathbb{R}^2,$$

with

$$W_1 = \begin{pmatrix} 1 & -1 \\ -1 & 2 \\ 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad b_1 = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \qquad W_2 = \begin{pmatrix} 1 & -1 & 2 & 1 \end{pmatrix}, \qquad b_2 = 0.$$

Each preactivation

$$\ell_1(x) = x_1 - x_2, \quad \ell_2(x) = -x_1 + 2x_2 + 1, \quad \ell_3(x) = 2x_1 + x_2 - 1, \quad \ell_4(x) = x_1 + 2x_2$$

defines a tropical hypersurface $\mathcal{T}(f_i) = \{\ell_i(x) = 0\}$, so the four hidden neurons carve the plane into polygonal regions. On any such region the activation vector

$$m(x) = (1_{\ell_1(x)>0}, \ldots, 1_{\ell_4(x)>0})$$

is constant and the gradient of the network is

$$\nabla\nu(x) = W_2 \ \mathrm{diag}(m(x))\, W_1 = \sum_{i=1}^{4} m_i(x)\, W_2(i,:) W_1(i,:).$$

Thus each of the 2^4 possible activation vectors determines a candidate gradient

$$g(m) = W_2\, \mathrm{diag}(m)\, W_1.$$

Not all activation vectors are feasible. A vector m is feasible only if there exists an x satisfying all inequalities $\ell_i(x) > 0$ for the units marked active and $\ell_i(x) \le 0$ for the units marked inactive. Feasibility means that the corresponding region in the hyperplane arrangement is nonempty, which also means that the cell dual to that sign pattern actually appears in the subdivision of the Newton polytope.

As an example, the activation pattern

$$m = (1, 1, 0, 1)$$

requires simultaneously

$$x_1 > x_2, \qquad x_1 \le \frac{1 - x_2}{2}, \qquad x_2 < 0, \qquad x_1 > -2x_2,$$

which are mutually inconsistent.

By contrast, the region where all four hidden units are active is feasible. On this region $m = (1, 1, 1, 1)$ and

$$\nabla\nu(x) = g(1, 1, 1, 1) = W_2 W_1.$$

A direct computation gives

$$W_2 W_1 = (7,\ 1),$$

so the gradient is constantly $(7, 1)$ throughout this region.

The hyperplanes $\ell_i(x) = 0$ are

$$\begin{aligned}
H_1 &: x_1 - x_2 = 0,\\
H_2 &: -x_1 + 2x_2 + 1 = 0,\\
H_3 &: 2x_1 + x_2 - 1 = 0,\\
H_4 &: x_1 + 2x_2 = 0.
\end{aligned}$$

These are precisely the tropical hypersurfaces $\mathcal{T}(f_i)$ of the hidden tropical polynomials. According to the statement above and the tropical description above, the gradient is constant on each connected component of the complement of $H_1 \cup \cdots \cup H_4$, and it may change only when crossing one of these tropical hypersurfaces.

In the accompanying Python notebook [4] I implement the network and compute the gradient. I then evaluate $\nabla\nu(x)$ on a grid in the square domain $[-2, 2]^2$, visualize the resulting vector field, and overlay the four hyperplanes H_i. Figure 1 shows the expected structure with each polygonal region having a constant gradient determined by its activation signature, and gradient changes occur exactly when a tropical hypersurface is crossed.

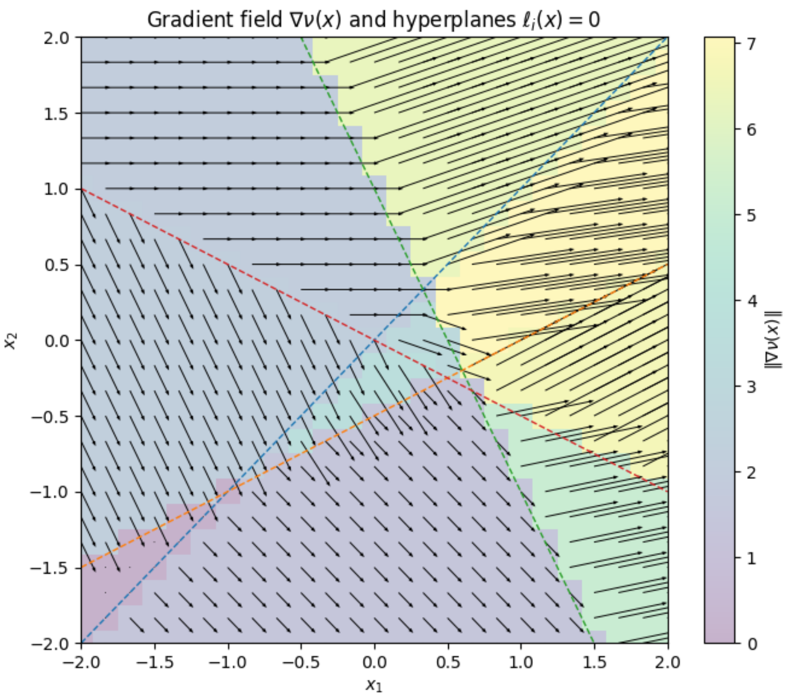

Figure 1: Gradient field $\nabla\nu(x)$ for the integer-weight network and the arrangement of hyperplanes $H_i = \{\ell_i(x) = 0\}$.

4 Discussion & Future Directions

4.1 Summary

The tropical viewpoint provides a geometric description of the piecewise-linear structure of ReLU networks with integer weights. Each hidden neuron defines a tropical polynomial with a tropical hypersurface given by the affine hyperplane where its ReLU activation switches. The arrangement of these tropical hypersurfaces partitions the input space into polyhedral regions on which the gradient of the network is constant. In this sense, the gradient field of a ReLU network is entirely encoded in the combinatorics of the underlying tropical hypersurfaces. The tropical interpretation also clarifies when activation patterns are feasible or infeasible, corresponding respectively to nonempty or empty polyhedral cells.

A further observation is that this geometric structure provides a natural notion of identifiability. Different choices of weights and biases can induce the same tropical hypersurface arrangement and therefore the same collection of linear regions and gradients. In that sense, the parameters themselves are not identifiable, but the induced tropical subdivision and gradient field are. One could explore further what assumptions or restrictions would lead to the identifiability of the parameters, and there exist literature in tropical geometry discussing this for more theoretical purposes that could possibly be applied and prove useful for neural networks.

4.2 Tropical geometry and universal approximation

In class we discussed the Universal Approximation Theorem for sigmoidal activation functions. We also discussed that any non-polynomial activation yields universal approximation on compact sets. Since the ReLU activation $\sigma(z) = \max\{z, 0\}$ is non-polynomial, ReLU networks are dense in $C(K)$ for every compact (closed and bounded in this case) $K \subset \mathbb{R}^d$. In this sense, one may think of a ReLU architecture as containing a very large family of piecewise-linear functions whose linear regions can be made arbitrarily fine.

4.3 Connections to tropical sigmoidal networks

The work of Chu and Kuo [3] revisits the Universal Approximation Theorem for sigmoidal networks through a tropical lens. They consider classical multi-layer perceptrons with sigmoidal activations of the form

$$G_N(x) = \sum_{j=1}^{N} \alpha_j \sigma(w_j^\top x + \theta_j),$$

where σ is sigmoidal, and show how to construct networks whose decision boundaries agree with prescribed geometric shapes at initialization. They design networks by composing sigmoidal approximations of half-space indicators arranged so that the resulting decision boundary follows a given polyhedral scheme.

In the ReLU setting, the hard max structure makes tropical rational functions an exact model. In the sigmoidal setting, the max is replaced by a smooth approximation, but the asymptotic limiting behavior of steep sigmoids exhibits tropical behavior. This connection suggests that tropical thinking can inform interpretability and initialization for a broader class of networks, not only ReLU.

Possible applications

One of the main assumptions made in the paper is that the weights are integers, which rarely occurs in the real world, and as such it might be useful to view the model as having gradients that approximate those of the networks we are trying to use and study. In real-world settings, such as prediction models or scoring systems in fields such as medicine or finance, where points lying well inside a tropical region are handled stably, since small perturbations do not change the gradient, where by contrast, points lying close to a hypersurface are inherently fragile, as arbitrarily small changes in certain directions can trigger a different response. The tropical structure therefore could provide some notion of where a model is locally stable, where it is highly sensitive, and where you can interpret abrupt changes due to the biases near the boundaries.

The choice of polynomial in the computation was also purposefully simple in order to be realistic to compute and help visually establish the connections drawn in this project. In practical settings where we have many more neurons and a much more complicated subdivision, some numerical techniques including sampling from certain regions or other estimation methods can be used to reconstruct the tropical subdivision from evaluations of the network.

References

[1] L. Zhang, G. Naitzat, and L.-H. Lim. Tropical geometry of deep neural networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018.

[2] D. Maclagan and B. Sturmfels. *Introduction to Tropical Geometry.* Graduate Studies in Mathematics, vol. 161, American Mathematical Society, 2015. Corresponding lectures available at `https://www.youtube.com/playlist?list=PLRy_Pn1LtSpfYE39TuCZrJAi7-0z44sQB`

[3] Y.-S. Chu and Y.-C. Kuo. From Universal Approximation Theorem to Tropical Geometry of Multi-Layer Perceptrons. arXiv:2510.15012, 2025.

[4] F. Doughan. Gradient fields and tropical hypersurfaces of a two-layer ReLU network. Colab notebook, 2025. Available at
`https://colab.research.google.com/drive/1ZWIXFqEDSUh5P1mdlxVC16Q0E6IQG1jY?usp=sharing`.

*The built-in Gemini tool was used to format the code and add documentation. Colab deletes chat history after each session, but the prompting was rudimentary—just instructing it to format the code nicely and add comments.

*Near the end of the project, I also prompted ChatGPT for feedback on my full project pdf according to the rubric as well as the code before submitting. It was a very back and forth conversation where I continuously supplied updated versions of this pdf asking for feedback each time. I tried to make sure to interpret its feedback in my own words after attempting to understand the points it was bringing up. Please find the transcript at the following link: `https://chatgpt.com/share/e/6938b1ef-6aa0-800d-a3c1-bd2b43bb996b`

Please let me know if you have trouble accessing any of the links.